

PGR106

Linear Algebra

Lecture 1 – Introduction & Vectors



Welcome!

- Welcome to PGR106.
- Instructor is Hadi Zahmatkesh.
- This is a **Linear Algebra** course with **Python**.
- Please ask questions/let me know if I am difficult to understand.
- 2 x 2 hours lectures + 2 x 2 hours exercise sessions.

Course Requirements & Assessment

- Submission of one assignment
- Final written exam

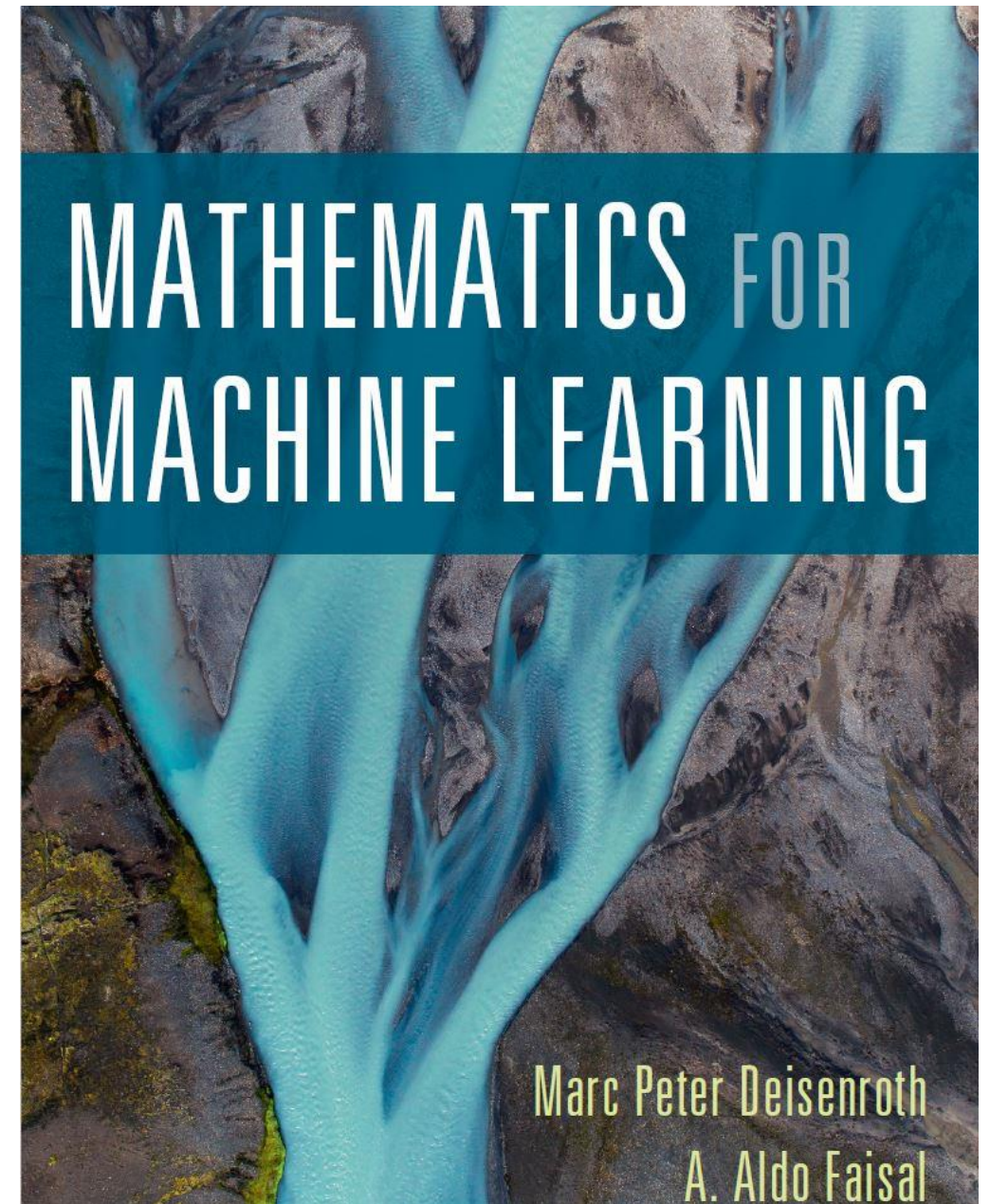


Course Materials & Project!

- All the slides that I will talk about during lectures will be up on Canvas before lectures.
- Project will be posted on Canvas too.
- Project will be done individually.
- During exercise sessions, you will practice what you learn in the lectures.

The Book!

- Mathematics for Machine Learning by Marc Peter Deisenroth and A. Aldo Faisal
- Can get it easily on Akademika bookstore.
- The PDF of the book is freely available at <https://mml-book.github.io/>



Getting Help

- You can always email me. Please have **PGR106** in the title.
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Outline

- Lecture 1 – Introduction & Vectors
- Lecture 2 – Matrices
- Lecture 3 – Matrix Determinant & Inverse
- Lecture 4 – System of Linear Equations
- Lecture 5 – Matrix Equations $Ax = b$
- Lecture 6 – Norms & Basis Vectors
- Lecture 7 – Vector Subspaces & Orthogonality
- Lecture 8 – Eigenvalues & Eigenvectors
- Lecture 9 – Singular Value Decomposition & Least Squares Problem
- Sample Exam Questions

What is Linear Algebra?

- Linear Algebra is a branch of **mathematics** that has a huge impact on **data science**.
- Linear Algebra helps **data scientist** to extract useful information from data.
- A good knowledge of Linear Algebra is essential for understanding **Machine Learning** (**ML**) techniques.

What is Linear Algebra?

- **Linear Algebra** is the study of **vectors**, **matrices**, and in general **tensors** and the rules to manipulate them.
 - ✓ Over the years, data is written in tables and spreadsheets
 - ✓ With advent of computers, data can be stored in **vectors**, **matrices**, and **tensors**
 - ✓ Operations on **vectors**, **matrices**, and **tensors** are done using **Linear Algebra**
- **Linear Algebra** is a way of representing data using **linear equations**.
- **Linear Algebra** behind **ML** helps in developing suitable solutions.

Why Linear Algebra?

- Linear Algebra is used in preprocessing & transformation of data
- Using Linear Algebra, data can be viewed in higher dimensions
- Linear Algebra is the basis of ML techniques (e.g., regression, optimization, classification, dimensionality reduction, etc.)
- Linear Algebra helps in understanding how ML models work

Terminology

- **Linear Algebra**: focus on theory and proofs
- **Matrix Algebra**: focus on applications

Linear Algebra Applications

- Data Analysis and Big Data
- Computer Graphics
 - ✓ An image that you can move around on the screen
- Graph Theory: a mathematical framework to understand networks
 - ✓ Air traffic control
 - ✓ Internet
 - ✓ Social media networks

Linear Algebra Applications

- Linear Algebra is used to solve problems in engineering, calculus, differential equations, finance, economics, biology, etc.
- Linear Algebra is the key backbone of most modern computational algorithms, and therefore learning linear algebra will improve your programming skills!

Scalars and Vectors

- **Scalar** is just a number.

For example,

7

$\frac{1}{4}$

2.86

In terms of data science, scalar is an object that has a magnitude

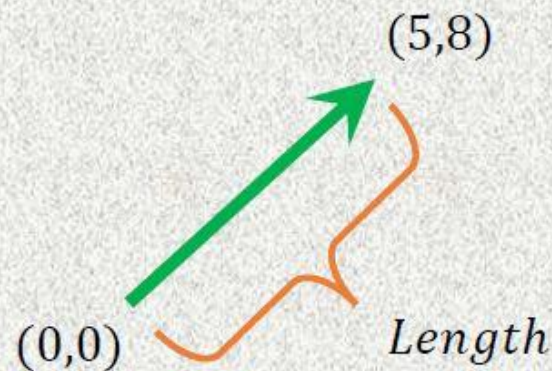
Age -> 37

Salary -> 7000

Scalars and Vectors

- **Vectors** are fundamental part of linear algebra and data science.
- A **vector** is an **ordered** collection of elements or numbers.

$$\begin{bmatrix} 5 \\ 8 \end{bmatrix}$$



Mathematics

$$\begin{bmatrix} \text{age} \\ \text{income} \\ \text{Expense} \end{bmatrix} \begin{bmatrix} 32 \\ 8000 \\ 5000 \end{bmatrix}$$

Data Science

Vectors

- **Vectors** are **1D** array of numbers, arranged in order, so elements can be accessed by its index.

$$\vec{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ \vdots \\ \vdots \\ \vdots \\ v_n \end{bmatrix}$$

Maximum Temperature (° C)	Minimum Temperature (° C)	Snow (cm)	Dew (°)	Feels Like (° C)	Humidity (%)	Wind Speed (kmph)
28	15	0	10	28	31	11

$$\vec{w} = \begin{bmatrix} 28 \\ 15 \\ 0 \\ 10 \\ 28 \\ 31 \\ 11 \end{bmatrix}$$

$$\vec{w} = [28 \quad 15 \quad 0 \quad 10 \quad 28 \quad 31 \quad 11]$$

Vectors

$$\vec{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

$$\vec{u} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

$$\vec{w} = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}$$

$$\vec{y} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\mathbb{R}^2$$

Vectors

$$\vec{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$$

$$\vec{u} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}$$

$$\vec{w} = \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix}$$

$$\vec{y} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$$

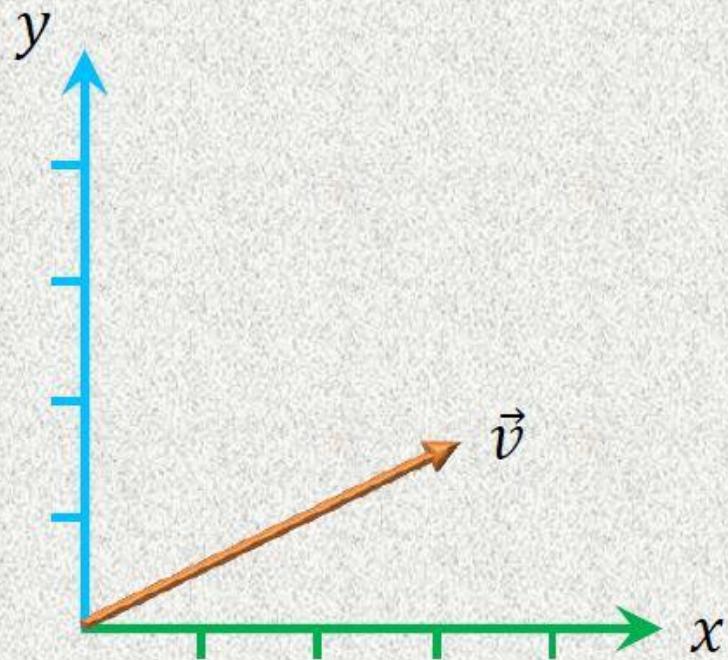
$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$\mathbb{R}^3$$

Vectors

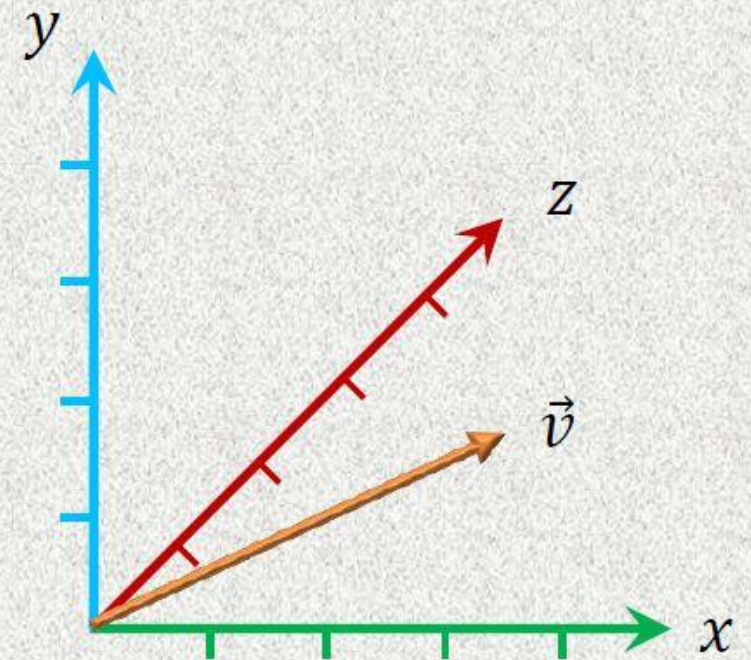
$\mathbb{R}^2$

$$\vec{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$



$\mathbb{R}^3$

$$\vec{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$$



Vectors

$$\vec{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ \vdots \\ \vdots \\ \vdots \\ v_n \end{bmatrix}$$

$$\vec{u} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ \vdots \\ \vdots \\ \vdots \\ u_n \end{bmatrix}$$

$$\vec{w} = \begin{bmatrix} w_1 \\ w_2 \\ w_3 \\ \vdots \\ \vdots \\ \vdots \\ w_n \end{bmatrix}$$

$$\mathbb{R}^n$$

Vectors in **2D** Space

- **Vectors** in $\mathbb{R}^2$

Direction

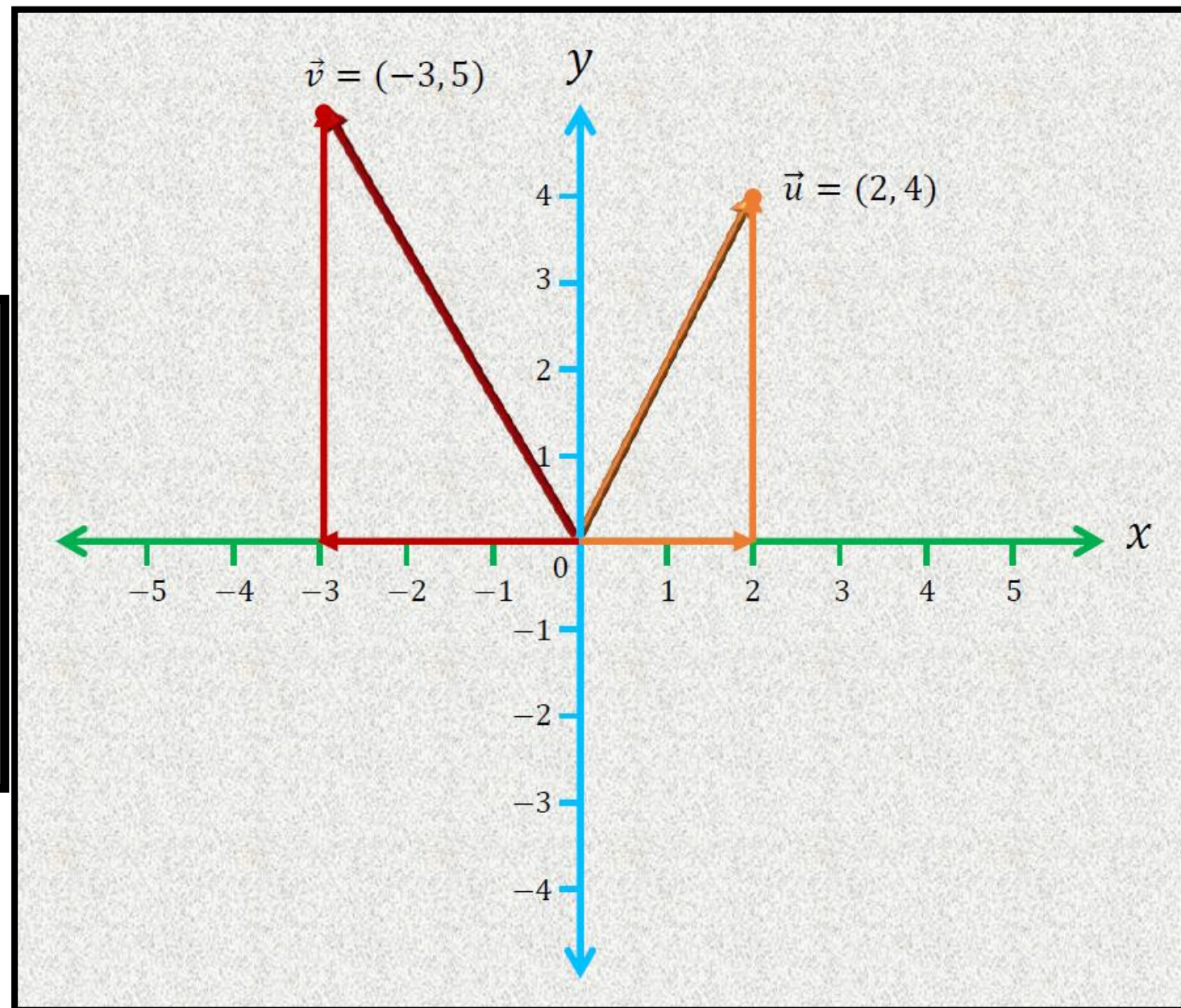
Magnitude

$$\vec{u} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$$

$$\vec{u} = \begin{bmatrix} 2 & 4 \end{bmatrix}$$

$$\vec{v} = \begin{bmatrix} -3 \\ 5 \end{bmatrix}$$

$\vec{v} = \begin{bmatrix} -3 & 5 \end{bmatrix}$



Vectors in 2D Space

- **Vectors** in $\mathbb{R}^2$

For example:

A person whose age is 32 years and income is \$8000 per month

$$\vec{P1} = \begin{bmatrix} 32 \\ 8000 \end{bmatrix} = \begin{bmatrix} 8000 \\ 32 \end{bmatrix}$$

A person whose age is 35 years and income is \$5000 per month

$$\vec{P2} = \begin{bmatrix} 35 \\ 5000 \end{bmatrix}$$

Vectors in **2D** Space

- **Vectors** in $\mathbb{R}^2$

$$\vec{v_1} = \begin{bmatrix} 5 \\ 8 \end{bmatrix}$$

$$\vec{v_2} = \begin{bmatrix} 8 \\ 5 \end{bmatrix}$$

$$\vec{v_3} = \begin{bmatrix} 5 \\ 8 \end{bmatrix}$$

$$\vec{v_1} \neq \vec{v_2}$$

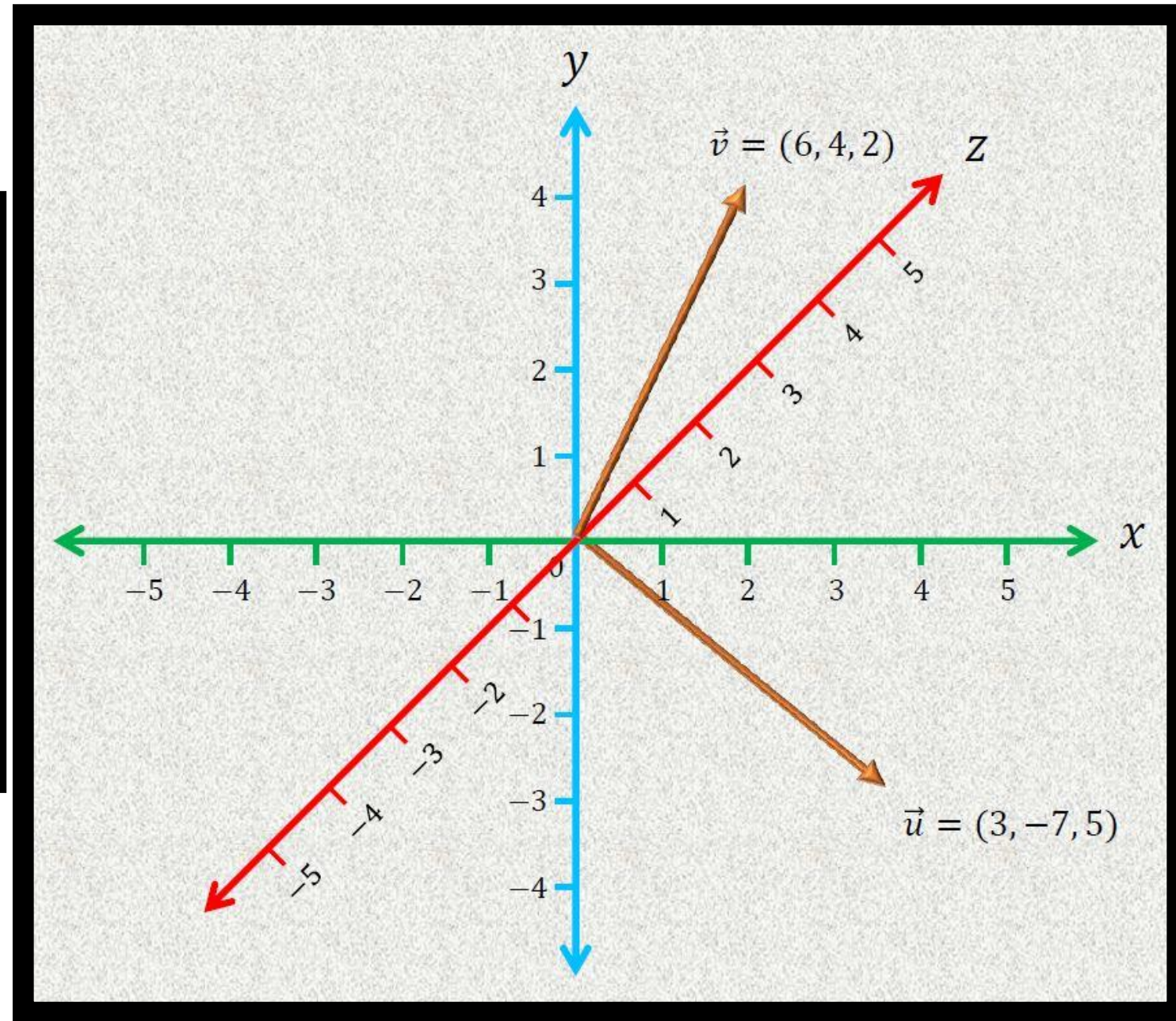
$$\vec{v_1} = \vec{v_3}$$

Vectors in **3D** Space

- **Vectors** in $\mathbb{R}^3$

$$\vec{u} = \begin{bmatrix} 3 \\ -7 \\ 5 \end{bmatrix} \quad \vec{u} = [3 \quad -7 \quad 5]$$

$$\vec{v} = \begin{bmatrix} 6 \\ 4 \\ 2 \end{bmatrix} \quad \vec{v} = [6 \quad 4 \quad 2]$$



Vectors in 3D Space

- **Vectors** in $\mathbb{R}^3$

For example:

A person whose age is 32 years, income is \$8000 per month and expenses are \$5000

$$\vec{P} = \begin{bmatrix} 32 \\ 8000 \\ 5000 \end{bmatrix} \in \mathbb{R}^3$$

Vectors in **nD** Space

- **Vectors** in $\mathbb{R}^n$

	Maximum Temperature (° C)	Minimum Temperature (° C)	Snow (cm)	Dew (°)	Feels Like (° C)	Humidity (%)	Visibility (km)	Wind Speed (kmph)
<i>London</i>	28	15	0	10	27	31	10	12
<i>Paris</i>	25	14	0	8	23	45	8	16

$$\vec{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ \vdots \\ \vdots \\ \vdots \\ v_n \end{bmatrix} \in \mathbb{R}^n$$

$$\vec{L} = \begin{bmatrix} 28 \\ 15 \\ 0 \\ 10 \\ 27 \\ 31 \\ 10 \\ 12 \end{bmatrix} \quad \vec{P} = \begin{bmatrix} 25 \\ 14 \\ 0 \\ 8 \\ 23 \\ 45 \\ 8 \\ 16 \end{bmatrix}$$

Vectors in **nD** Space

- **Vectors** in $\mathbb{R}^n$

	Maximum Temperature (° C)	Minimum Temperature (° C)	Snow (cm)	Dew (°)	Feels Like (° C)	Humidity (%)	Visibility (km)	Wind Speed (kmph)
<i>London</i>	28	15	0	10	27	31	10	12
<i>Paris</i>	25	14	0	8	23	45	8	16

$$\vec{L} = \begin{bmatrix} 28 \\ 15 \\ 0 \\ 10 \\ 27 \\ 31 \\ 10 \\ 12 \end{bmatrix}$$

$$\vec{P} = \begin{bmatrix} 25 \\ 14 \\ 0 \\ 8 \\ 23 \\ 45 \\ 8 \\ 16 \end{bmatrix}$$

$$\begin{aligned} \mathbb{R}^2 &\rightarrow 2D \\ \mathbb{R}^3 &\rightarrow 3D \\ \mathbb{R}^n &\rightarrow nD \end{aligned}$$

Vectors in **nD** Space

- **Data Science** deals with data in higher dimensions
- Properties of **vectors** in $\mathbf{R}^n$ are the same as properties of **vectors** in $\mathbf{R}^2$ and $\mathbf{R}^3$

	Maximum Temperature (° C)	Minimum Temperature (° C)	Snow (cm)	Dew (°)	Feels Like (° C)	Humidity (%)	Visibility (km)	Wind Speed (kmph)
<i>London</i>	28	15	0	10	27	31	10	12
<i>Paris</i>	25	14	0	8	23	45	8	16

$$\vec{L} = \begin{bmatrix} 28 \\ 15 \\ 0 \\ 10 \\ 27 \\ 31 \\ 10 \\ 12 \end{bmatrix}$$

$$\vec{P} = \begin{bmatrix} 25 \\ 14 \\ 0 \\ 8 \\ 23 \\ 45 \\ 8 \\ 16 \end{bmatrix}$$

Column Vectors

Zero Vectors

- A vector with all the components as **zeros** is called the **Zero Vector**.

$$\mathbb{R}^2$$

$$\vec{u} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\mathbb{R}^3$$

$$\vec{v} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\mathbb{R}^n$$

$$\vec{v} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ \vdots \\ 0 \end{bmatrix}$$

Ones Vectors

- A vector with all the components as **ones** is called the **Ones Vector** ($\mathbf{1}_n$).

$\mathbb{R}^2$

$$\vec{u} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \mathbf{1}_2$$

$\mathbb{R}^3$

$$\vec{v} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \mathbf{1}_3$$

$\mathbb{R}^n$

$$\vec{v} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ \vdots \\ \vdots \\ 1 \end{bmatrix} = \mathbf{1}_n$$

Operations on Vectors

- **Addition**: is performed by adding the corresponding components of the vectors.
- **Subtraction**: is also like performing addition of vectors.

$$\vec{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

$$\vec{u} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

$$\vec{v} + \vec{u} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} + \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} v_1 + u_1 \\ v_2 + u_2 \end{bmatrix}$$

Operations on Vectors

- Addition

For example,

*A family where the husband's income is \$8000
and his expenses are \$5000 per month*

$$\vec{h} = \begin{bmatrix} 8000 \\ 5000 \end{bmatrix}$$

*And wife's income is \$6000
and her expenses are \$4000 per month*

$$\vec{w} = \begin{bmatrix} 6000 \\ 4000 \end{bmatrix}$$

What is the family's income and expenses per month?

$$\vec{h} + \vec{w} = \begin{bmatrix} 8000 \\ 5000 \end{bmatrix} + \begin{bmatrix} 6000 \\ 4000 \end{bmatrix} = \begin{bmatrix} 8000 + 6000 \\ 5000 + 4000 \end{bmatrix} = \begin{bmatrix} 14000 \\ 9000 \end{bmatrix}$$

Operations on Vectors

- Addition

$$\mathbb{R}^2$$

$$\vec{u} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

$$\vec{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

$$\vec{u} + \vec{v} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} + \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} u_1 + v_1 \\ u_2 + v_2 \end{bmatrix}$$

Operations on Vectors

- Addition

$\mathbb{R}^3$

$$\vec{u} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}$$

$$\vec{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$$

$$\vec{u} + \vec{v} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} + \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} u_1 + v_1 \\ u_2 + v_2 \\ u_3 + v_3 \end{bmatrix}$$

Operations on Vectors

- Addition

$$\mathbb{R}^n$$
$$\vec{u} + \vec{v} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ \vdots \\ u_n \end{bmatrix} + \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ \vdots \\ v_n \end{bmatrix} = \begin{bmatrix} u_1 + v_1 \\ u_2 + v_2 \\ u_3 + v_3 \\ \vdots \\ u_n + v_n \end{bmatrix}$$

Operations on Vectors

- Addition

$$\vec{u} = \begin{bmatrix} 3 \\ -7 \\ 5 \end{bmatrix} \quad \vec{v} = \begin{bmatrix} 6 \\ 4 \\ 2 \end{bmatrix}$$

$$\vec{u} + \vec{v} = \begin{bmatrix} 3 \\ -7 \\ 5 \end{bmatrix} + \begin{bmatrix} 6 \\ 4 \\ 2 \end{bmatrix} = \begin{bmatrix} 3 + 6 \\ -7 + 4 \\ 5 + 2 \end{bmatrix} = \begin{bmatrix} 9 \\ -3 \\ 7 \end{bmatrix}$$

Operations on Vectors

- Scalar Multiplication

$$\vec{u} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ \vdots \\ \vdots \\ \vdots \\ u_n \end{bmatrix} \quad c\vec{u} = c * \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ \vdots \\ \vdots \\ \vdots \\ u_n \end{bmatrix} = \begin{bmatrix} c * u_1 \\ c * u_2 \\ c * u_3 \\ \vdots \\ \vdots \\ \vdots \\ c * u_n \end{bmatrix}$$

Operations on Vectors

- Scalar Multiplication

For example,

*A person whose income per month is \$8000
and his expenses are \$5000 per month*

$$\vec{h} = \begin{bmatrix} 8000 \\ 5000 \end{bmatrix}$$

What is the person's income and expenses for 3 months?

$$3 \vec{p} = 3 * \begin{bmatrix} 8000 \\ 5000 \end{bmatrix} = \begin{bmatrix} 3 * 8000 \\ 3 * 5000 \end{bmatrix} = \begin{bmatrix} 24000 \\ 15000 \end{bmatrix}$$

Operations on Vectors

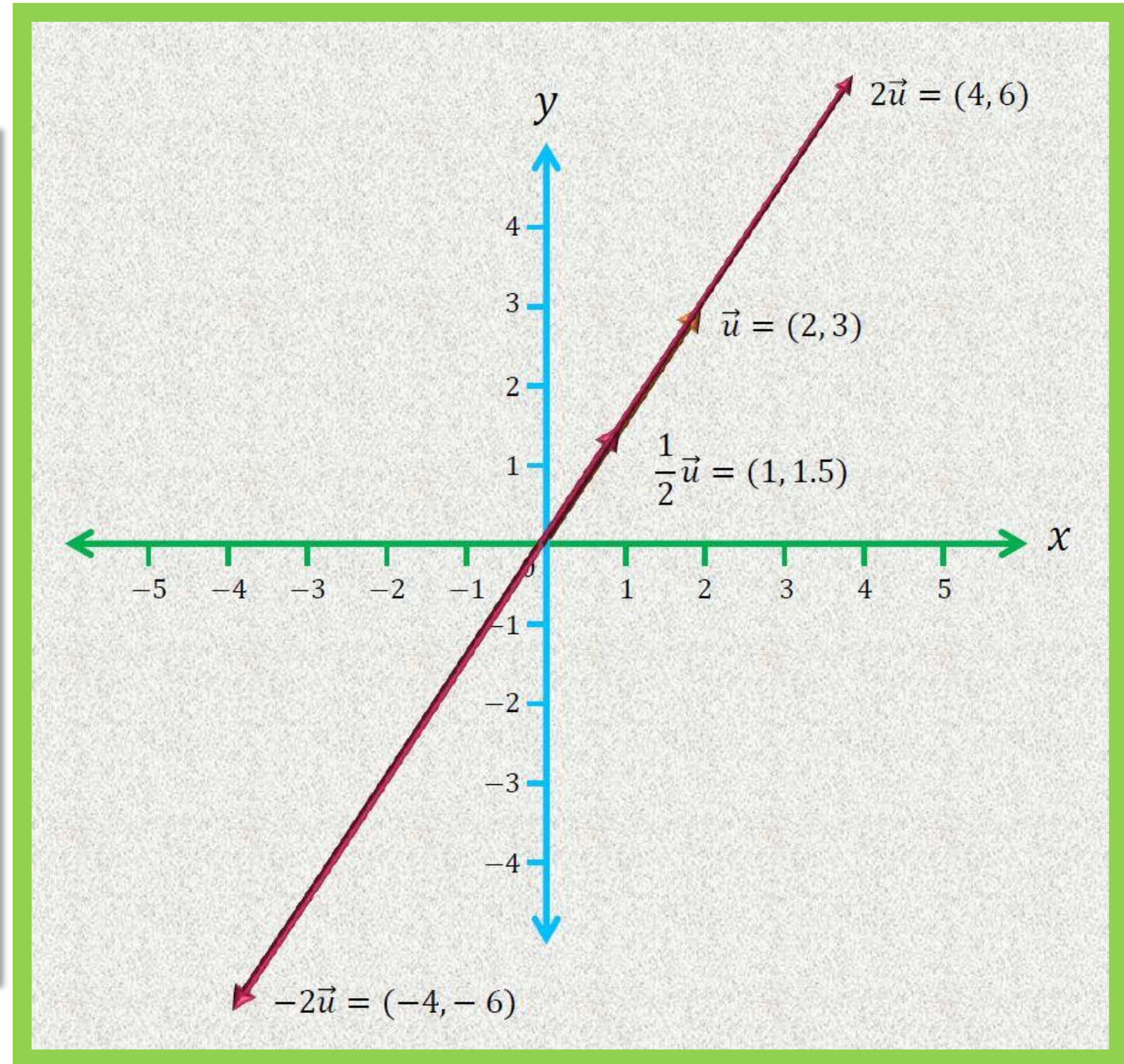
- Scalar Multiplication

$$\vec{u} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

$$2\vec{u} = 2 * \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 4 \\ 6 \end{bmatrix}$$

$$0.5 \vec{u} = 0.5 * \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 \\ 1.5 \end{bmatrix}$$

$$-2\vec{u} = -2 * \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} -4 \\ -6 \end{bmatrix}$$



Operations on Vectors

- Scalar Multiplication

$$\vec{u} = \begin{bmatrix} 3 \\ -7 \\ 5 \end{bmatrix}$$

$$2\vec{u} = 2 * \begin{bmatrix} 3 \\ -7 \\ 5 \end{bmatrix} = \begin{bmatrix} 2 * 3 \\ 2 * -7 \\ 2 * 5 \end{bmatrix} = \begin{bmatrix} 6 \\ -14 \\ 10 \end{bmatrix}$$

$$-3\vec{u} = -3 * \begin{bmatrix} 3 \\ -7 \\ 5 \end{bmatrix} = \begin{bmatrix} -3 * 3 \\ -3 * -7 \\ -3 * 5 \end{bmatrix} = \begin{bmatrix} -9 \\ 21 \\ -15 \end{bmatrix}$$

Vector Properties

$$\vec{u} + \vec{v} = \vec{v} + \vec{u}$$

$$(\vec{u} + \vec{v}) + \vec{w} = \vec{u} + (\vec{v} + \vec{w})$$

$$\vec{u} + \mathbf{0} = \vec{u}$$

Vector Properties

$$c(\vec{u} + \vec{v}) = c\vec{u} + c\vec{v}$$

$$(c + d)\vec{u} = c\vec{u} + d\vec{u}$$

$$c(d\vec{u}) = (cd)\vec{u}$$

Linear Combination of Vectors

- Linear Combination: is a combination of addition and scalar multiplication operations on vectors.
- The heart of linear algebra is in the combination of addition and scalar multiplication operations.

$$\vec{u} = \begin{bmatrix} 3 \\ -7 \end{bmatrix} \qquad \vec{v} = \begin{bmatrix} 6 \\ 4 \end{bmatrix}$$

$$\begin{aligned} 2\vec{u} + 3\vec{v} &= 2 * \begin{bmatrix} 3 \\ -7 \end{bmatrix} + 3 * \begin{bmatrix} 6 \\ 4 \end{bmatrix} \\ &= \begin{bmatrix} 6 \\ -14 \end{bmatrix} + \begin{bmatrix} 18 \\ 12 \end{bmatrix} = \begin{bmatrix} 6 + 18 \\ -14 + 12 \end{bmatrix} = \begin{bmatrix} 24 \\ -2 \end{bmatrix} \end{aligned}$$

$$\vec{x} = c_1 \vec{v}_1 + c_2 \vec{v}_2 + c_3 \vec{v}_3 + \dots \dots + c_n \vec{v}_n$$

Vector Transpose

	Maximum Temperature (° C)	Minimum Temperature (° C)	Snow (cm)	Dew (°)	Feels Like (° C)	Humidity (%)	Visibility (km)	Wind Speed (kmph)
London	28	15	0	10	27	31	10	12
Paris	25	14	0	8	23	45	8	16

$$\vec{L} = \begin{bmatrix} 28 \\ 15 \\ 0 \\ 10 \\ 27 \\ 31 \\ 10 \\ 12 \end{bmatrix} \text{ Column Vector}$$

$$\vec{P} = \begin{bmatrix} 25 \\ 14 \\ 0 \\ 8 \\ 23 \\ 45 \\ 8 \\ 16 \end{bmatrix} \text{ Column Vector}$$

$$\vec{L} = [28 \quad 15 \quad 0 \quad 10 \quad 27 \quad 31 \quad 10 \quad 12]$$

Row Vector

$$\vec{P} = [25 \quad 14 \quad 0 \quad 8 \quad 23 \quad 45 \quad 8 \quad 16]$$

Row Vector

Vector Transpose

- Transpose: is turning a column vector into a row vector and vice versa.

$$\vec{L} = \begin{bmatrix} 28 \\ 15 \\ 0 \\ 10 \\ 27 \\ 31 \\ 10 \\ 12 \end{bmatrix}$$

$$\vec{L}^T = [28 \quad 15 \quad 0 \quad 10 \quad 27 \quad 31 \quad 10 \quad 12]$$

$$\vec{L}^{TT} = \vec{L}$$

Dot Product or Inner Product

- Dot Product: is the sum of product of the corresponding components of each vector.
- Dot Product of two vectors determines how similar the two vectors are.

$$\vec{u} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} \quad \vec{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

$$\vec{u} \cdot \vec{v} = u_1 * v_1 + u_2 * v_2$$

Dot Product or Inner Product

- Dot Product: is also defined as the **matrix product**.

$$\vec{u} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} \quad \vec{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

$$\vec{u} \cdot \vec{v} = u^T v = \begin{bmatrix} u_1 & u_2 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

$$\vec{u} \cdot \vec{v} = u^T v = u_1 * v_1 + u_2 * v_2$$

Dot Product or Inner Product

$$\vec{u} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} \quad \vec{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$$

$$\vec{u} \cdot \vec{v} = u^T v = \begin{bmatrix} u_1 & u_2 & u_3 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$$

$$\vec{u} \cdot \vec{v} = u_1 * v_1 + u_2 * v_2 + u_3 * v_3$$

Dot Product or Inner Product

$$\vec{u} = \begin{bmatrix} u_1 \\ u_2 \\ \cdot \\ \cdot \\ u_n \end{bmatrix} \quad \vec{v} = \begin{bmatrix} v_1 \\ v_2 \\ \cdot \\ \cdot \\ v_n \end{bmatrix} \quad \vec{u} \cdot \vec{v} = u^T v = [u_1 \quad u_2 \quad \cdot \quad \cdot \quad u_n] \begin{bmatrix} v_1 \\ v_2 \\ \cdot \\ \cdot \\ v_n \end{bmatrix}$$

$$\vec{u} \cdot \vec{v} = u_1 * v_1 + u_2 * v_2 + \cdots \dots \dots + u_n * v_n$$

Dot Product or Inner Product

$$\vec{u} = \begin{bmatrix} 2 \\ 4 \end{bmatrix} \quad \vec{v} = \begin{bmatrix} -3 \\ 5 \end{bmatrix}$$

$$\vec{u} \cdot \vec{v} = u^T v = \begin{bmatrix} 2 & 4 \end{bmatrix} \begin{bmatrix} -3 \\ 5 \end{bmatrix}$$

$$\vec{u} \cdot \vec{v} = 2 * -3 + 4 * 5$$

$$\vec{u} \cdot \vec{v} = -6 + 20 = 14$$

Dot Product or Inner Product

$$\vec{u} = \begin{bmatrix} 3 \\ -7 \\ 5 \end{bmatrix} \quad \vec{v} = \begin{bmatrix} 6 \\ 4 \\ 2 \end{bmatrix} \quad \vec{u} \cdot \vec{v} = u^T v = \begin{bmatrix} 3 & -7 & 5 \end{bmatrix} \begin{bmatrix} 6 \\ 4 \\ 2 \end{bmatrix}$$

$$\vec{u} \cdot \vec{v} = 3 * 6 + (-7) * 4 + 5 * 2$$

$$\vec{u} \cdot \vec{v} = 18 - 28 + 10$$

$$\vec{u} \cdot \vec{v} = 0$$

Dot Product or Inner Product

- Dot Product: is the sum of product of the corresponding components of each vector.

Positive Dot Product: Vectors point in a generally **similar** direction
(The angle between the vectors is between 0 and 90)

Negative Dot Product: Vectors point in a generally **opposite** direction
(The angle between the vectors is between 90 and 180)

Zero Dot Product: Vectors are **perpendicular** to each other
(The angle between the vectors is 90)

Outer Product

- Outer Product: is a product of a column vector with a row vector.

$$\vec{u} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} \quad \vec{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

$$\vec{u} \otimes \vec{v} = u v^T = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} [v_1 \quad v_2]$$

$$\vec{u} \otimes \vec{v} = u v^T = \begin{bmatrix} u_1 * v_1 & u_1 * v_2 \\ u_2 * v_1 & u_2 * v_2 \end{bmatrix}$$

Outer Product

- Outer Product: is a product of a column vector with a row vector.

$$\vec{u} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} \quad \vec{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$$

$$\vec{u} \otimes \vec{v} = u v^T = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} [v_1 \quad v_2 \quad v_3]$$

$$\vec{u} \otimes \vec{v} = u v^T = \begin{bmatrix} u_1 * v_1 & u_1 * v_2 & u_1 * v_3 \\ u_2 * v_1 & u_2 * v_2 & u_2 * v_3 \\ u_3 * v_1 & u_3 * v_2 & u_3 * v_3 \end{bmatrix}$$

Outer Product

- Outer Product: is a product of a column vector with a row vector.

$$\vec{u} = \begin{bmatrix} u_1 \\ u_2 \\ \cdot \\ \cdot \\ u_n \end{bmatrix}$$

$$\vec{v} = \begin{bmatrix} v_1 \\ v_2 \\ \cdot \\ \cdot \\ v_n \end{bmatrix}$$

$$\vec{u} \otimes \vec{v} = u v^T = \begin{bmatrix} u_1 \\ u_2 \\ \cdot \\ \cdot \\ u_n \end{bmatrix} \begin{bmatrix} v_1 & v_2 & \cdot & \cdot & v_n \end{bmatrix}$$

Outer Product

$$\vec{u} = \begin{bmatrix} 2 \\ 4 \end{bmatrix} \quad \vec{v} = \begin{bmatrix} -3 \\ 5 \end{bmatrix}$$

$$\vec{u} \otimes \vec{v} = u v^T = \begin{bmatrix} 2 \\ 4 \end{bmatrix} \begin{bmatrix} -3 & 5 \end{bmatrix}$$

$$\vec{u} \otimes \vec{v} = u v^T = \begin{bmatrix} 2 * -3 & 2 * 5 \\ 4 * -3 & 4 * 5 \end{bmatrix} = \begin{bmatrix} -6 & 10 \\ -12 & 20 \end{bmatrix}$$

Outer Product

$$\vec{u} = \begin{bmatrix} 3 \\ -2 \\ 5 \end{bmatrix} \quad \vec{v} = \begin{bmatrix} 6 \\ 4 \\ 2 \end{bmatrix}$$

$$\vec{u} \otimes \vec{v} = u v^T = \begin{bmatrix} 3 \\ -2 \\ 5 \end{bmatrix} [6 \quad 4 \quad 2]$$

$$\vec{u} \otimes \vec{v} = u v^T = \begin{bmatrix} 3 * 6 & 3 * 4 & 3 * 2 \\ -2 * 6 & -2 * 4 & -2 * 2 \\ 5 * 6 & 5 * 4 & 5 * 2 \end{bmatrix} = \begin{bmatrix} 18 & 12 & 6 \\ -12 & -8 & -4 \\ 30 & 20 & 10 \end{bmatrix}$$

Inner Product and Outer Product

$$\vec{u} = \begin{bmatrix} u_1 \\ u_2 \\ \cdot \\ \cdot \\ u_n \end{bmatrix} \quad \vec{v} = \begin{bmatrix} v_1 \\ v_2 \\ \cdot \\ \cdot \\ v_n \end{bmatrix}$$

1 Dimensions of vectors must be same

2 Result of Inner Product or Dot Product: Scalar Value

$\vec{u} \cdot \vec{v} \rightarrow \text{Scalar Value}$

3 Result of Outer Product: Matrix

$\vec{u} \otimes \vec{v} \rightarrow n \times n \text{ Matrix}$

Inner Product and Outer Product



Dot Product or Inner Product:

$$\vec{u} \cdot \vec{v} = u^T v$$



Outer Product:

$$\vec{u} \otimes \vec{v} = u v^T$$

End of Lecture 1



MATHEMATICS FOR MACHINE LEARNING

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